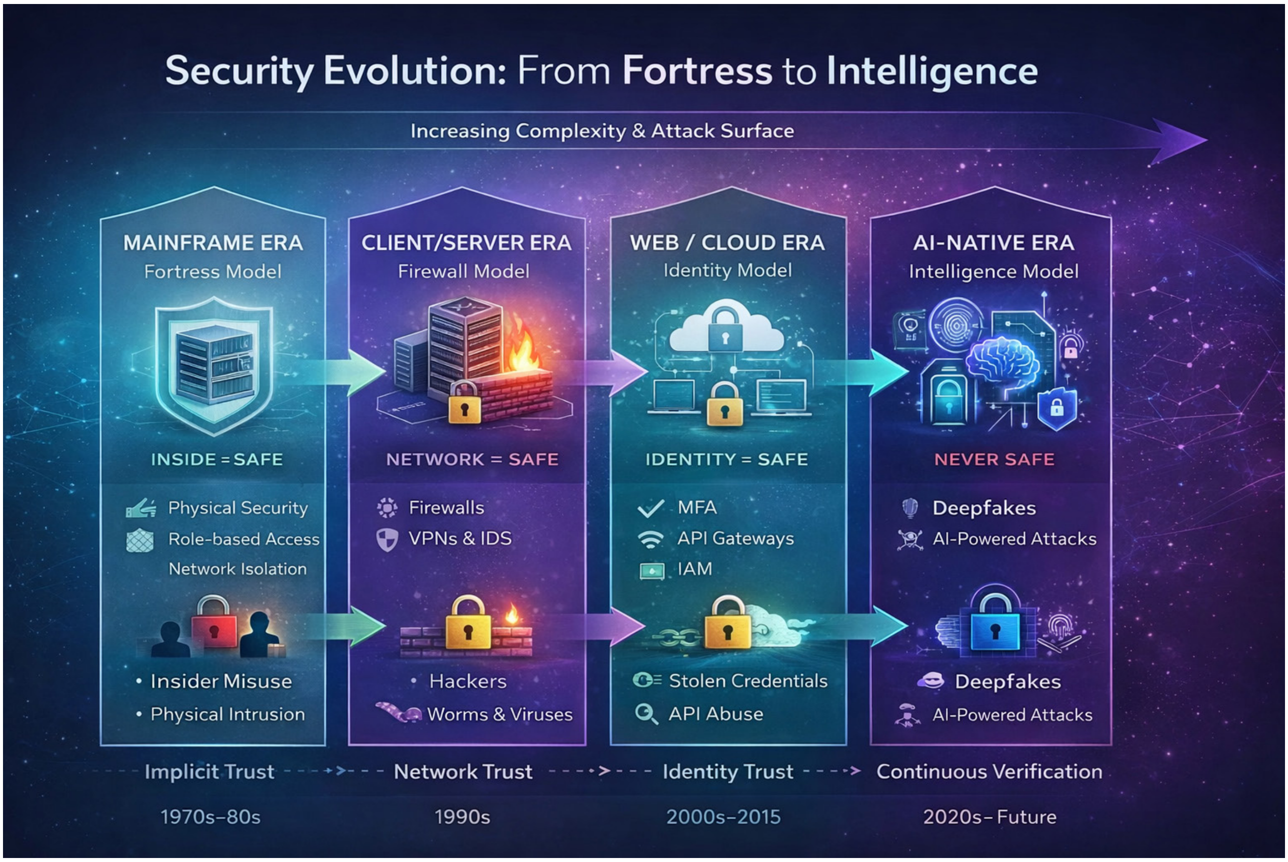

Figure 1: Security evolution (Image source: AI generated)

- *Supply chain exposure:* Compromised software updates or vendor tools can introduce threats into otherwise isolated systems.
- *Maintenance backdoors:* Temporary network bridges created for troubleshooting or patching often outlive their intended purpose.
- *Configuration drift:* Over time, exceptions accumulate, and 'temporary access' quietly becomes permanent exposure.

The shift from deterministic to probabilistic AI models is creating a significant calculability gap. In traditional systems, outcomes are 'calculable' because they follow fixed, binary rules—press a button, and you receive the same outcome every time. AI, however, operates within a spectrum of probabilities. While it may not produce the exact same outcome twice, it must be governed to operate within strictly defined 'allowable ranges' of safety and performance. This loss of mathematical certainty creates a gap where risk becomes harder to predict. To mitigate this, we must adopt a multi-model validation strategy and integrate Human-in-the-Loop (HITL) oversight, ensuring that human judgment remains the final anchor when AI approaches the boundaries of those allowable ranges.

Overall, the gap between IT and OT has narrowed significantly. Any compromise in one domain can now move laterally to impact the other, creating a bidirectional risk to the entire enterprise.

Mitigation strategies

Historically, our focus was on preventing physical intrusion through site security. This evolved into securing the network perimeter and, eventually, shifted towards identity as the primary barrier. In today's AI-driven landscape, Zero Trust has become the essential mantra we all rely on to navigate these complexities.

Today, the attack surface has expanded significantly with the integration of IoT and OT into traditional IT environments. This shift has escalated the potential impact from mere data theft or ransomware to threats against human life, particularly when critical infrastructure—such as power grids, medical equipment, transportation systems, and public utilities—is compromised.

To address this, we need a paradigm shift in awareness. Vigilance can no longer be the responsibility of a few specialists; it must scale to the population level. While awareness/training is essential, we face a persistent hurdle: complacency. Security is often treated as a checkbox for regulation or compliance rather than a core priority. Unfortunately, many organisations only act when they are forced to, rather than adopting a truly proactive, institutionalised security-first mindset.